Cross-Domain Statistical Analysis for Robust DoS Detection in Heterogeneous Network Environments*

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Abstract

Network Intrusion Detection Systems (NIDS) often suffer from severe performance degradation in unseen environments due to domain shift across datasets. In this work, we investigate whether common statistical characteristics of Denial-of-Service (DoS) traffic can be identified across heterogeneous NIDS datasets. We unified flow-level statistical features and analyzed five public intrusion detection datasets through Kernel Density Estimation (KDE)-based distribution analysis. Our observations revealed significant distributional differences in conventional flow features, indicating strong dataset dependency. To better capture intrinsic attack behaviors, we additionally designed TCP flag ratio-based features using SYN and ACK packet statistics.

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1 Introduction

Network Intrusion Detection Systems (NIDS) often experience severe performance degradation when applied to unseen environments due to domain shift across datasets. Variations in traffic distributions, network configurations, and attack scenario generation methodologies make it difficult for machine learning-based NIDS to generalize across heterogeneous environments. This limitation becomes increasingly important in dynamic E-Business infrastructures where intrusion detection systems must operate under continuously changing traffic conditions.

In this work, we investigate whether common statistical characteristics of Denial-of-Service (DoS) traffic can be identified across multiple NIDS datasets. Rather than focusing on dataset-specific optimization, our study aims to explore domain-invariant features that consistently distinguish benign and DoS traffic across heterogeneous network environments.

2 Methodology

To ensure feature consistency, we unified flow-level statistical features using CICFlowMeter and analyzed five publicly available intrusion detection datasets: Improved CIC-IDS2018, CIC-IoT2023, OD-IDS2022, CIC-ToN-IoT, and USB-IDS-1. Initial experiments showed that directly applying common statistical flow features across datasets produced limited performance, indicating significant distributional differences among datasets.

To better understand these differences, we conducted feature-wise distribution analysis using Kernel Density Estimation (KDE). Specifically, we analyzed the distributions of Length of Fwd Packet, Length of Bwd Packet, Average Packet Size, Flow Duration, Flow Bytes/s, and Flow Packets/s for benign and DoS traffic.

3 Analysis and Conclusion

Among the selected features, Flow Packets/s exhibited distinguishable patterns for several DoS subclasses across multiple datasets as shown in **Figure 1**, suggesting its potential as a transferable feature for cross-domain intrusion detection. Also, we explored TCP flag-based ratio features designed to capture intrinsic behavioral characteristics of DoS attacks. Since DoS traffic often generates abnormal TCP control packet patterns, we derived several ratio-based features using SYN and ACK flag statistics, including ACK_SYN_Ratio and ACK_TotalPkt_Ratio. Preliminary analysis on CIC-IDS2018 and USB-IDS-1 demonstrated that these features exhibit meaningful separability between benign and DoS traffic. KDE visualizations further suggested that these ratio-based features preserve relatively consistent patterns across datasets, as shown in **Figure 2**.

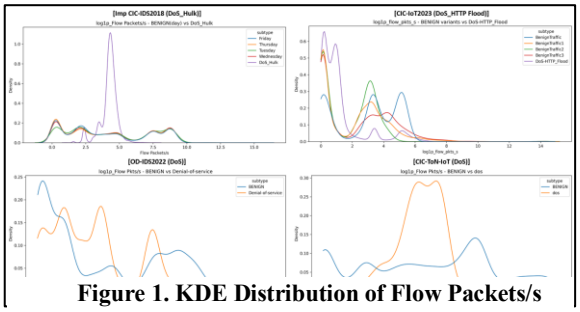


Figure 1. KDE Distribution of Flow Packets/s

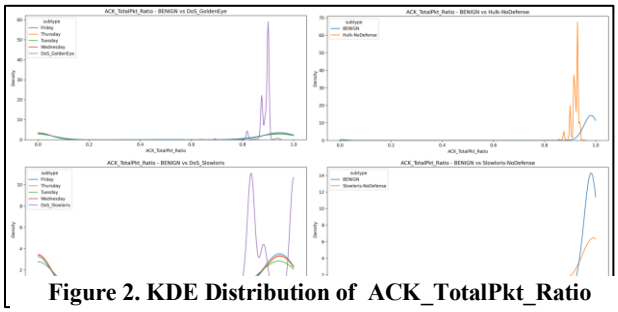


Figure 2. KDE Distribution of ACK_TotalPkt_Ratio

Our analysis revealed that conventional flow-level statistical features exhibit significant distributional differences across datasets, limiting their effectiveness for cross-domain intrusion detection. Among the analyzed features, Flow Packets/s showed relatively consistent patterns for several DoS subclasses across multiple datasets. In addition, protocol-level TCP flag ratio features, particularly ACK_SYN_Ratio and ACK_TotalPkt_Ratio, demonstrated meaningful separability between benign and DoS traffic in both Improved CIC-IDS2018 and USB-IDS-1. However, the observed TCP flag ratio patterns were not fully consistent across all DoS subclasses. In particular, the DoS Slowloris subclass showed opposite distributional tendencies between USB-IDS-1 and CIC-IDS2018, indicating that protocol-level features may still be affected by dataset-specific attack implementations. Therefore, further analysis is required to determine which TCP flag ratio features can be considered truly transferable across datasets.

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